Roll No. I Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-2)

PHARMACEUTICAL ORGANIC CHEMISTRY-I

Subject Code : BP-202T

M.Code : 74968

Date of Examination : 09-05-2026

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Briefly write about the following:

a) Why are alkanes referred to as "paraffins"?

b) Draw the structure of Hexamine and state its medicinal use.

c) Compare the boiling points of n-pentane and neopentane. Explain the difference.

d) Define hyperconjugation with a suitable example.

e) Why is trichloroacetic acid more acidic than acetic acid?

f) Identify the monomeric units of Paraldehyde.

g) What is the "peroxide effect" in the addition of HBr to propene?

h) Arrange the following in increasing order of SN2 reactivity: 1-bromobutane, 2-bromobutane, 2-bromo-2-methylpropane.

i) Explain why benzyl alcohol does not undergo the same oxidation reactions as aliphatic alcohols.

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1) State the use of Dimethyl phthalate in pharmacy.

SECTION – B

2. Discuss the kinetics, stereochemistry, and orientation (Saytzeff's rule) of E1 and E2 reactions. Explain why secondary alkyl halides can undergo both.

3. Explain the mechanism of Aldol Condensation and Crossed Aldol Condensation. Why do aldehydes without α-hydrogens undergo the Cannizzaro reaction instead?

4. Compare SN1 and SN2 mechanisms. Discuss how the nature of the solvent and the structure of the substrate dictate which pathway is followed.

SECTION – C

5. Explain the Diels-Alder reaction using 1,3-butadiene and ethene as examples.

6. Discuss the Inductive Effect and its role in determining the basicity of aliphatic amines.

7. Describe the qualitative tests used to identify Carboxylic Acids and Amides.

8. Write the structure and specific pharmaceutical uses of Chloral Hydrate and Iodoform.

9. Explain sp^3 hybridization in methane with a diagram of orbital overlapping.

10 What is Allylic Rearrangement? Explain with the example of a conjugated diene.

11. Give the structure, IUPAC name, and uses of Lactic acid and Citric acid.

12. Illustrate Structural Isomerism in organic compounds with five carbons (C_5H_{12}).

13. Explain the mechanism and importance of Ozonolysis in locating double bonds.

N.B.E : Disclosure of Identity by writing Mobile No. or Making of passive request on any page of Answer Sheet will lead to UMC against the Student.

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